AMAGI Master Glossary v1.0

Canonical Terminology Register for the AMAGI Architectural Class

VERSION 1.0 — OCTOBER 2026

> Single normative source of terminology for all documents of the AMAGI Architectural Class. Where the local acronym table or term usage of any corpus document conflicts with this register, this register governs. Statuses: **CANONICAL**, **DEPRECATED** (canonical replacement specified; use prohibited in new and revised documents), **RELATED** (external ecosystem; not part of the AMAGI canon).

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1. Purpose and Authority

This glossary establishes the canonical terminology of the AMAGI Architectural Class and resolves the acronym collisions and term drift identified across the corpus during the 2026 consistency freeze. The corpus has accumulated up to four coexisting expansions for single acronyms (MAG, ATL), three for TTP, and two for MPE and CRTM; uncontrolled term variance is a defect class of its own, independently of any numerical parameter conflicts, because it obscures whether documents describe the same or different architectural roles.

Authority rules are stated in §9. In summary: this register governs terminology; each document remains authoritative for its own technical content; naming corrections propagate at the next scheduled version of each affected document (correction vehicles listed per entry in §7) and do not alter architectural roles, interfaces, or assigned responsibilities unless enacted through a separate ADR.

2. Normative References

- Amagi Open Specification v1.1 (DOI: 10.5281/zenodo.20753403), §2 acronym table, Appendix A — primary canon for component terms;
- Amagi Framework v2.0 (DOI: 10.5281/zenodo.20753400) — architectural canon, ADR register, PQC suite;
- Amagi-NIPU v1.1 (DOI: 10.5281/zenodo.20753401), §2 invariant realization, §5 FSM chain-reset;
- Amagi-MEM v1.1 (DOI: 10.5281/zenodo.20753402), §3.3 ATL — TTP Engine and Audit Diode, §3.5 PQC;
- Amagi-FMEDA v1.0 (DOI: 10.5281/zenodo.20753410) — canonical usage «Tautological Truth Predicate»; v1.1 supersedes for quantitative content;
- NIST FIPS 203, 204, 205; NIST SP 800-208; IEC 61508 series (terminology: SIL, HFT, PFDavg, SFF, Type A/B, proof test);
- Master Parameter Sheet v2.1 (Wave 1) — layer model, instance register RI-1/RI-2, T1/T_M;
- Master Version Control Document v1.1 (Wave 2) — document statuses and correction wave plan.

3. Terminology Governance

Status	Meaning	Rule
CANONICAL	Approved expansion/term	Mandatory in all new and revised documents
DEPRECATED	Withdrawn expansion/term	Shall not be introduced; existing occurrences corrected at the next version of the affected document (vehicle per §7)
RELATED	Term of an adjacent ecosystem or of the external literature	May appear only with explicit disambiguation at first use

Term additions and status changes enter into force exclusively through a version bump of this glossary. Substantive architectural changes (roles, interfaces, responsibilities) are out of scope of this register and proceed through the ADR mechanism of the Open Specification.

4. Canonical Terms — Architecture (Layer A)

Term	Expansion	Definition (condensed)	Canon fixed in
AMAGI	— (proper name)	Architectural class for hardware-enforced AI safety systems defined by six invariants and the TTP Triad	Open Spec v1.1 §2
Layer A	Architecture layer	Substrate-independent canon of the class: INV-001…006, SCC protocol semantics, TTP predicate, MAG target structure, isolation profiles, integrity capability formula	MPS v2.1 §1.1
Layer I	Implementation layer	Measured parameters of concrete implementation instances; append-only register; instances are never merged or averaged	MPS v2.1 §1.1
INV-001	Physical Non-Bypassability	Enforcement path cannot be bypassed at physical level	NIPU v1.1 §2
INV-002	Irreducible TCB	Minimal TCB, ≤500 lines HDL	NIPU v1.1 §2
INV-003	HW Separation of Control and Cognition	Control plane separated from cognitive (AI) plane in hardware	NIPU v1.1 §2
INV-004	Controlled Mutability of Critical Policies	Policy changes only via governed update path; active policy store immutable in Normal operation	NIPU v1.1 §2; update path per ADR-002
INV-005	Complete Mediation	All security-relevant operations mediated; hardware-enforced TTP chain	NIPU v1.1 §2
INV-006	Complete Audit Integrity	Audit trail complete and tamper-evident	NIPU v1.1 §2
Non-droppable invariants	—	INV-001, INV-005, INV-006 (irreducible safety core); INV-002/003/004 waivable per IAA v1.1 §4 matrix	Charter v1.2; Open Spec v1.1
SCC	Security Council Core	Minimal TCB enforcing TTP at processor level; sole trusted control plane	Open Spec v1.1 §8.1
TTP	**Tautological Truth Predicate**	Formal predicate enforced by the SCC: a configurable delay must elapse and the predicate must hold before activation is possible	FMEDA v1.0 §5; AMAGI TCAD manuscript
Temporal Triad	—	Temporal structure Trigger → Programmed Delay → Localized Activation; established as ADR-001	Open Spec v1.1; AF v2.0
NIPU	Neuro-Immune Processing Unit	Processor-level instantiation of the AMAGI class	NIPU v1.1 §1

Term	Expansion	Definition (condensed)	Canon fixed in
MEM	Self-Protecting, Context-Aware Memory Architecture	Memory-level instantiation of the AMAGI class; short form «Self-Protecting Memory Architecture» acceptable	MEM v1.1 title; Open Spec v1.1 §2
MPE	Memory Policy Enforcer	Validates all memory access requests against MAG policy	Open Spec v1.1 §2; MEM v1.1 §3.1
MAG	Memory Access Graph	128×128 eFuse/OTP matrix storing 4-bit policy codes governing all data movement	Open Spec v1.1 §2
ATL	Access Trigger Layer	Implements the TTP delay at memory level (0.5 ns–1 µs) and hosts the audit transaction logging function	Open Spec v1.1 §2; MEM v1.1 §3.3
Audit Diode	—	Unidirectional audit-logging hardware function embedded in the ATL (not delegated to SHMF)	MEM v1.1 §3.3
Audit Transaction Logger	—	ATL sub-function producing the tamper-evident audit trail (INV-006)	AF v2.0 §5; TM v1.0 SR-08
SHMF	Self-Healing Memory Fabric	ECC and hash-chain-based data integrity with self-isolation	Open Spec v1.1 §2
EGW	Execution Gateway	Hardware AND-gate in Secure World gating EXE cluster activation	Open Spec v1.1 §2; ADR-006
EXE	Execution cluster	Autonomous execution cluster; demoted to Normal World per ADR-006	Open Spec v1.1 §8
CRTM	Core Root of Trust for Measurement	Mask-ROM component authenticating SCC firmware at boot; TCB member per ADR-007	Open Spec v1.1 §2
AFC	Analog Fallback Circuit	Hardware safety auxiliary with separate LDO	Open Spec v1.1 §2
DDL	Digital Delay Lock	PVT-calibrated delay element of the TTP delay chain	FMEDA v1.0 §5
Amagi-Q	— (product designation)	Shared quantum-resistant cryptographic engine (ML-DSA-44, FIPS 204)	Open Spec v1.1 §8; NIST Profile v1.0
Dual-Tier Trust Model	—	Two mutually exclusive deployment tiers; resolves INV-004 tension; ADR-002; Tier 2 PCCP-compliant OTA	Open Spec v1.1; AF v2.0
Isolation profiles S / D	—	Profile S: subsystem-level isolation (reference, current); Profile D: device-level isolation (target, Phase 2)	MPS v2.1 §1.3
PTCB	Policy-Enforcing Trusted Computing Base	Collective designation of the enforcement TCB (SCC + MPE + MAG + ATL + SHMF + CRTM + Amagi-Q)	MPS v2.1 §20 (new term, introduced in MPS v2.1)
RI-1	Reference Implementation 1	VHDL historical prototype (benchtop measurements)	MPS v2.1 §1.2
RI-2	Reference Implementation 2	SystemVerilog **current reference implementation** per the AMAGI TCAD manuscript	MPS v2.1 §1.2; VC v1.1 §6

5. Canonical Terms — Assurance, Verification, and Documents

Document set (canonical titles):

Designation	Canonical title	Record
AF	Amagi Framework Specification (v2.0: Protection Profile + Integration Specification)	10.5281/zenodo.20753400

Designation	Canonical title	Record
AOS / Open Spec	Amagi Open Specification (v1.1 current)	10.5281/zenodo.20753403
IAA	Amagi Architectural Class Definition (Invariants and Architecture) v1.1	10.5281/zenodo.20145198
NIPU / MEM	Component specifications Amagi-NIPU, Amagi-MEM	10.5281/zenodo.20753401 / .20753402
HARA	Hazard Analysis and Risk Assessment	10.5281/zenodo.20753408
FMEDA	Failure Modes, Effects, and Diagnostic Analysis	10.5281/zenodo.20753410
TM (TRACE)	Bidirectional Hazard-to-Verification Traceability Matrix	10.5281/zenodo.20753412
NIST-PROF	AMAGI NIST AI RMF Profile v1.1	10.5281/zenodo.20753404
MPS	Master Parameter Sheet	10.5281/zenodo.20145212
VC	Master Version Control Document	10.5281/zenodo.20145223
S-APL	Strict Architectural Principle License	10.5281/zenodo.20753406
Charter	AGIAM Founding Charter	10.5281/zenodo.20753872
TS-NOT	AMAGI Framework Trade Secrets Notice	10.5281/zenodo.17886699
ADR	Architectural Decision Record (ADR-001…008 ratified; register in Open Spec v1.1 Table 12)	—
Glossary	AMAGI Master Glossary (this document)	[new record]

Assurance terminology (IEC 61508 frame): SIL (Safety Integrity Level); 1oo1 / 1oo2 (one-out-of-one / one-out-of-two architectures); HFT (Hardware Fault Tolerance); β (common-cause failure factor); PFDavg (average Probability of dangerous Failure on Demand); SFF (Safe Failure Fraction); λ_DU (dangerous undetected failure rate; FIT = failures per 10^9 hours); Type A / Type B subsystem (IEC 61508-2; AMAGI TCB: Type B); **T1** — proof test interval, 8 760 h; **T_M** — mission/design life, 87 600 h (distinction normative per MPS v2.1 §3.3); Certain Event (software/AI failure classification, P≈1, excluded from the safety-critical path per INV-003+INV-005, FMEDA v1.0); PVT (Process-Voltage-Temperature).

Integrity Capability Formula — canonical statement of achievable SIL per deployment configuration (SIL 2 @1oo1 current; SIL 3 @1oo2, β=0.10, Phase 2 target 2027); normative in MPS v2.1 §3.1, referenced by HARA v1.1 and FMEDA v1.1.

Verification terminology: SPIN / Promela (model checker / modelling language); LTL (Linear Temporal Logic); CEX (counterexample); property classes **SCC-S01…S09** (safety properties, PASS), **SCC-S10…S11** (architectural assumptions satisfied by hardware construction), **SCC-L01…L04** (liveness properties; CEX expected in unconstrained environment); weak fairness; fairness assumptions FA1/FA2; FSM chain-reset (reset-propagation logic refined in NIPU v1.1 §5, closing the latch-up trap identified by SPIN); latch-up (FSM stall condition, hardware vector closed in NIPU v1.1); Proof-of-Logic (prototype evidence scope: logic verified in SPIN, single XC7A35T device, post-synthesis timing evidence per Vivado 2023.2).

6. Canonical Terms — Cryptography

Term	Status	Definition / rule	Canon fixed in
ML-KEM-1024	CANONICAL	Key establishment, FIPS 203	Open Spec v1.1 §9; MEM v1.1 §3.5
ML-DSA-44	CANONICAL	Signature, FIPS 204; firmware and audit-log signing	Open Spec v1.1 §8–9; MEM v1.1 §3.5

Term	Status	Definition / rule	Canon fixed in
ML-DSA-87	CANONICAL	Signature, FIPS 204; CNSA 2.0 target option	Open Spec v1.1 §9
SLH-DSA-SHA2-128f	CANONICAL	Auxiliary hash/signature, FIPS 205; **stateless** hash-based digital signature; signature ~7 856 B	AF v2.0 §(resources); MEM v1.1 §3.5
HMAC-SHA256	CANONICAL	Current MAC (Kyber/HMAC row superseded by FIPS naming)	MEM v1.1 §3.5
«Stateful hash-based signatures»	SCOPED	SP 800-208 family (XMSS/LMS) is a **distinct, legitimately stateful** family; where firmware signing per SP 800-208 is meant, the scheme shall be named explicitly; classifying SLH-DSA (FIPS 205) as stateful is prohibited (§7, D-04)	NIST SP 800-208; FIPS 205
eFuse / OTP	CANONICAL	One-time-programmable policy storage of MAG	Open Spec v1.1 §2

7. Deprecated Terms (normative register)

Use of the following terms in new or revised documents is prohibited. Each entry lists the canonical replacement, the document fixing the canon, observed occurrences in the existing corpus, and the correction vehicle. Existing published records are immutable on Zenodo; corrections propagate through new versions per VC v1.1 §8.

#	Deprecated term	Canonical replacement	Canon fixed in	Observed occurrences (corpus)	Correction vehicle
D-01	**Mutable Action Gate**	Memory Access Graph (MAG)	Open Spec v1.1 §2; AF v2.0	HARA v1.0 §2 (component enumeration)	HARA v1.1
D-02	**Actuation Time Lock**	Access Trigger Layer (ATL)	Open Spec v1.1 §2; MEM v1.1 §3.3	FMEDA v1.0 §3, §5 (TTP delay chain description)	FMEDA v1.1
D-03	**SPHINCS+-256s** (incl. signature size 29 792 B; «SLH-DSA SHA2-256s»)	**SLH-DSA-SHA2-128f** (FIPS 205), signature ~7 856 B	AF v2.0 §(resources); MEM v1.1 §3.5	MPS v2.0 §4.1, §19	MPS v2.1 §4.1
D-04	**«Stateful hash-based signature(s)»** as classification of SLH-DSA/FIPS 205	Stateless (FIPS 205 defines stateless hash-based signatures; standard title cited in MPS v2.0 bibliography)	FIPS 205; Open Spec v1.1 §(normative references)	MPS v2.0 §20 (glossary row «SPHINCS+ — Stateful…»)	MPS v2.1 §20; scope rule in §6 of this glossary
D-05	Memory Architecture Grid	Memory Access Graph (MAG)	Open Spec v1.1 §2	MPS v2.0 §6, §20	MPS v2.1 §6, §20
D-06	Mutable Authenticated Graph	Memory Access Graph (MAG)	Open Spec v1.1 §2	FMEDA v1.0 §6 (component grouping)	FMEDA v1.1
D-07	Audit Trail Log (as component name)	Access Trigger Layer (ATL); audit function embedded in ATL as Audit Diode + Audit Transaction Logger	MEM v1.1 §3.3	HARA v1.0 §2	HARA v1.1 (rename only; architectural roles unchanged)
D-08	Access Time Latency	Access Trigger Layer (ATL)	Open Spec v1.1 §2	MPS v2.0 §20	MPS v2.1 §20

#	Deprecated term	Canonical replacement	Canon fixed in	Observed occurrences (corpus)	Correction vehicle
D-09	Trigger-Programmable Delay-Localized Activation (as TTP expansion)	TTP = **Tautological Truth Predicate**; the temporal structure is named **Temporal Triad**	FMEDA v1.0 §5; MPS v2.1 §2	Open Spec v1.1 §2 (acronym table)	Open Spec v1.2 corrigendum (Wave 3); until then §9/R1 of this glossary governs
D-10	Temporal Triad Parameters (as TTP expansion)	TTP = Tautological Truth Predicate	FMEDA v1.0 §5	MPS v2.0 §2, §20	MPS v2.1 §2, §20
D-11	Memory Protection Engine	Memory Policy Enforcer (MPE)	Open Spec v1.1 §2; MEM v1.1 §3.1	HARA v1.0 §2	HARA v1.1
D-12	Code Root of Trust Mask	Core Root of Trust for Measurement (CRTM)	Open Spec v1.1 §2 (TCG-consistent term)	FMEDA v1.0 §6; MPS v2.1 draft §20 (fixed by C13)	FMEDA v1.1; MPS v2.1 §20
D-13	Kyber; Dilithium	ML-KEM-1024 (FIPS 203); ML-DSA (FIPS 204)	FIPS 203/204; MEM v1.1 §3.5 update note	MPS v2.0 §4.1, §19 (legacy names)	MPS v2.1 §4
D-14	Artix-7 100T (as RI-2 device claim); 1.5 W (as RI-2 power claim)	XC7A35T; < 1.3 W (RI-2, per TCAD manuscript and AF v2.0 §9)	AF v2.0 §9	MPS v2.0 §5, §13 (instance-mixing)	MPS v2.1 §5.2 (per-instance split); family-level «Xilinx Artix-7» references remain valid
D-15	«TI = 10 years» (proof-test-interval wording for the 10-year figure)	T1 = 8 760 h (proof test interval); T_M = 87 600 h (mission/design life)	MPS v2.1 §3.3	AF v2.0 (lifecycle wording); FMEDA v1.0/MPS v2.0 «TI = 1 year» rows are T1-correct but mislabelled in comparisons	AF v2.1 corrigendum (Wave 3); FMEDA v1.1 §7.1
D-16	«Verified» as unconditional branding of parameter values	«Evidence» column with mandatory instance/source attribution	MPS v2.1 §11 (C8 rule)	MPS v2.0 (Verification column)	MPS v2.1 §5.2, §18

8. Acronym Collisions

This section states, for each colliding acronym, all expansions observed in the corpus and the single canonical resolution. The resolution applies to the whole corpus immediately upon publication of this glossary (precedence rule R1).

8.1 ATL — four expansions observed

Expansion	Documents	Resolution
Access Trigger Layer	Open Spec v1.1 §2; MEM v1.0 §3.3; MEM v1.1 §3.3; AF v2.0 §5	**CANONICAL**
Audit Trail Log	HARA v1.0 §2	DEPRECATED (D-07): rename only; the audit function is embedded **in** the ATL (Audit Diode, Audit Transaction Logger per MEM v1.1 §3.3) and is not a separate component, nor is it delegated to SHMF
Access Time Latency	MPS v2.0 §20	DEPRECATED (D-08)

Expansion	Documents	Resolution
Actuation Time Lock	FMEDA v1.0 §3, §5	DEPRECATED (D-02)

Resolution rule: in all documents ATL = Access Trigger Layer. Where the audit function is meant, the full phrases «Audit Diode» or «Audit Transaction Logger» are used. HARA v1.1 and FMEDA v1.1 apply the renames **without any change of architectural roles** (author decision of 2026-10, recorded as CCR-01 in VC v1.1 §7).

8.2 MAG — four expansions observed

Expansion	Documents	Resolution
Memory Access Graph	Open Spec v1.1 §2; MEM v1.0 §3.2; MEM v1.1 §3.2; AF v2.0; NIST Profile v1.0; Charter v1.0/1.1; AF v1.3	**CANONICAL**
Memory Architecture Grid	MPS v2.0 §6, §20	DEPRECATED (D-05)
Mutable Authenticated Graph	FMEDA v1.0 §6	DEPRECATED (D-06)
Mutable Action Gate	HARA v1.0 §2	DEPRECATED (D-01)

8.3 TTP — three in-corpus expansions, one external

Expansion	Documents	Resolution
Tautological Truth Predicate	FMEDA v1.0 §5; AMAGI TCAD manuscript	**CANONICAL**
Trigger-Programmable Delay-Localized Activation	Open Spec v1.1 §2 (acronym table)	DEPRECATED (D-09): the triad structure is named Temporal Triad; TTP denotes the predicate
Temporal Triad Parameters	MPS v2.0 §2, §20	DEPRECATED (D-10)
Temporally Programmed Delivery	Althea bioengineering corpus (CRISPR-TTP manuscript V3.0; Strategic Partnership Offer v1.0)	**RELATED** — separate ecosystem; retained there; any cross-ecosystem document shall disambiguate explicitly

External-literature caution: in security literature TTP commonly expands to Trusted Third Party. AMAGI documents using TTP in an IEC 61508 or security context shall define the term at first use.

8.4 CRTM — two expansions observed

Expansion	Documents	Resolution
Core Root of Trust for Measurement	Open Spec v1.1 §2	**CANONICAL**
Code Root of Trust Mask	FMEDA v1.0 §6; MPS v2.1 draft §20 (corrected, C13)	DEPRECATED (D-12)

8.5 MPE — two expansions observed

Expansion	Documents	Resolution
Memory Policy Enforcer	Open Spec v1.1 §2; MEM v1.0 §3.1; MEM v1.1 §3.1; AF v2.0	**CANONICAL**
Memory Protection Engine	HARA v1.0 §2	DEPRECATED (D-11)

8.6 MEM — component name vs record family

The designation MEM names both the memory-level component/instantiation and the Amagi-MEM document record family. Full canonical title: «Self-Protecting, Context-Aware Memory Architecture» (MEM v1.1 title page). The Open Spec v1.1 short form «Self-Protecting Memory Architecture» is an acceptable abbreviation. No other expansion is permitted.

8.7 SCC — no in-corpus variants

SCC = Security Council Core uniformly (IAA v1.0/v1.1; AF v1.4; Open Spec v1.1; NIST Profile v1.0). No deprecated variant exists; the expansion shall nevertheless be stated at first use in every document.

9. Precedence and Conflict-Resolution Rules

- **R1.** For terminology, this glossary prevails over the local acronym tables of any corpus document, including documents published simultaneously with or later than this glossary. Local acronym tables are informative.
- **R2.** Deprecated terms (§7) shall not be introduced in any new or revised document.
- **R3.** Existing occurrences are corrected at the next scheduled version of the affected document (vehicle per §7); interim citations of archived documents are annotated «terminology superseded by Master Glossary v1.0».
- **R4.** Naming corrections do not modify architectural roles, interfaces, or responsibilities (CCR-01, VC v1.1 §7). Any such modification requires an ADR and is out of scope of this register.
- **R5.** New terms enter the canon only through a version bump of this glossary; documents shall not introduce private expansions of shared acronyms.
- **R6.** A parameter without a layer tag (Layer A / Layer I) or an instance tag (RI-1 / RI-2) is void for consistency purposes (MPS v2.1 §1.1).

10. Change Log

Version	Date	Author	Changes
1.0	2026-10-26	A.M. Burlai (consistency freeze)	Initial register: 30+ canonical terms; 16 deprecated entries (D-01…D-16); 7 collision resolutions; precedence rules R1–R6